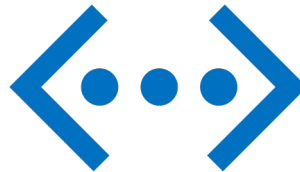


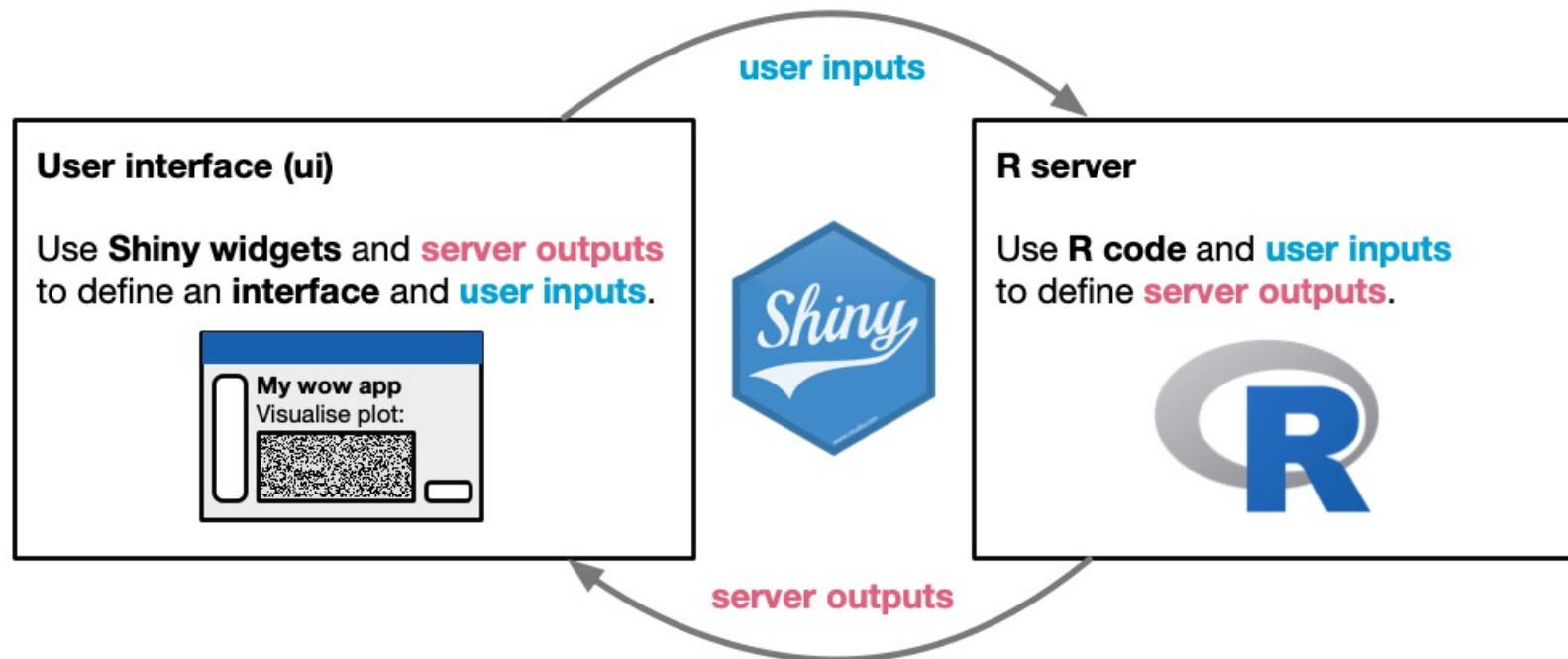
Shiny app deployment

Nanocourse Lecture 4, 8/5/25

Scott Saunders

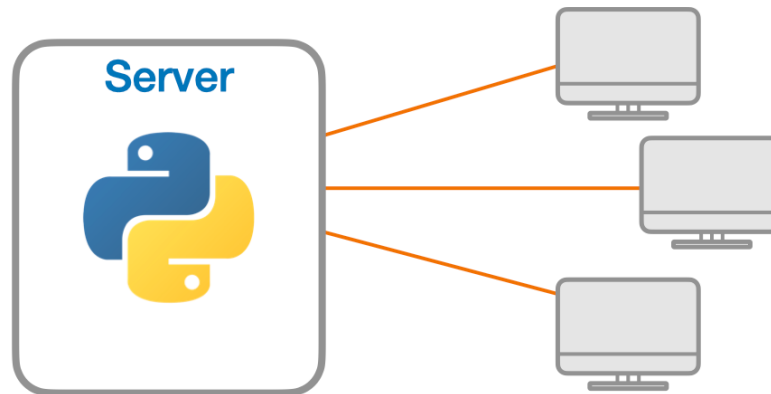


Shiny = server + interface



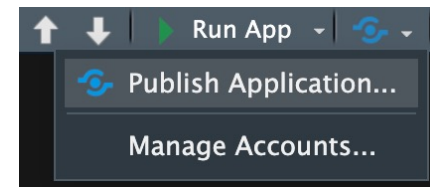
Option 1: Run a shiny server

**Traditional
Shiny
deployment**



Code exercise: deploy an app to shinyapps.io!

1. Make shinyapps account
2. Choose an app example
3. Press “publish” in RStudio
4. Test app load time & functionality on web
5. Checkout dashboard



Option 1: Run a shiny server (Shinyapps.io)

Pros

Free

Easy

Professionally maintained

Cons

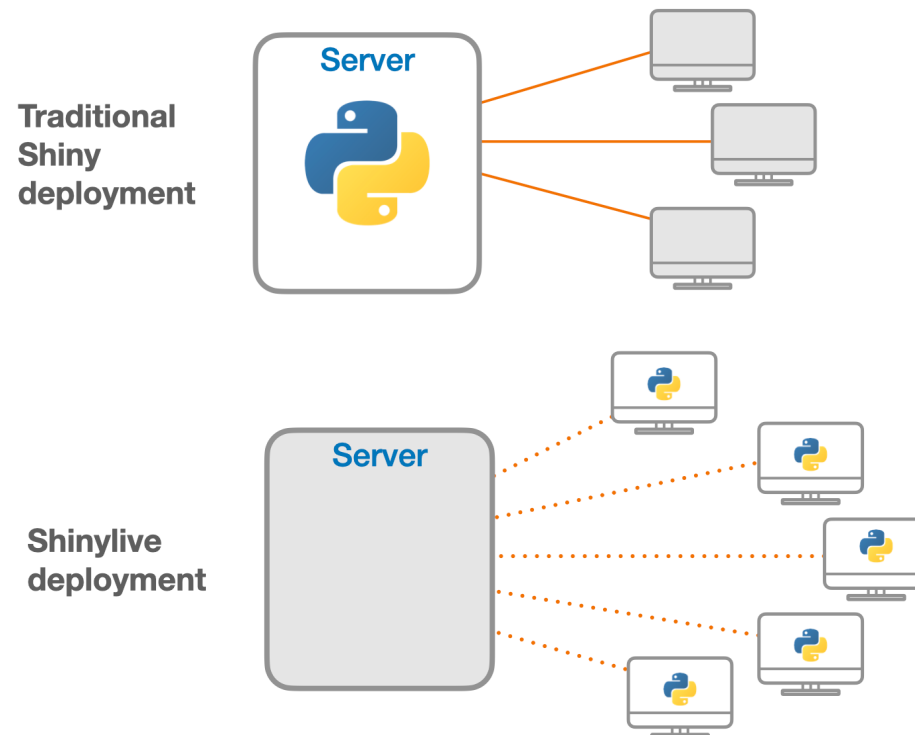
Limited apps

Limited users

Pay for advanced features



Option 2: Run shinylive (client side server)

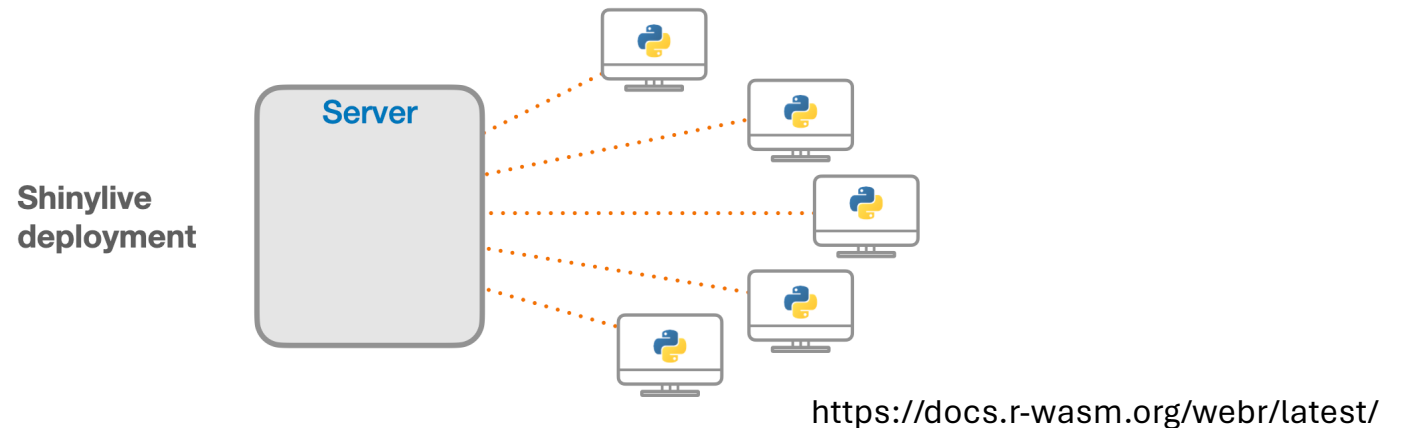


<https://shiny.posit.co/py/get-started/shinylive.html>

<https://posit-dev.github.io/r-shinylive/>

Option 2: Run shinylive (client side server)

- Static server gives user files
- User installs R / packages on the fly to create their own “server”
- Enabled by R that runs in web browser (webR)



Code exercise: deploy app to github pages with shinylive!

```
# R

install.packages("shinylive")

library("shinylive")
Library("httpuv")

export("my_app", "my_app_shinylive")

runStaticServer("my_app_shinylive")
```

- Github: Make new shiny course repo
- Make sure github pages is enabled for repo
- Upload or push shinylive app folder
- Go to URL!



Option 2: Run shinylive (client side server)

Pros

- Free
- Scalable
- Very simple “server”

Cons

- Slow startup
- Some limited functionality
- Users can see all source code



How can we easily share private data w/ collaborators?

- App password (shinyapps.io)
 - Shinymanager
 - Shinyauthr
- Shinylive (or server) + file upload = no sensitive data stored
- Share app files securely + user runs Rstudio / shiny. Install on lab computer...
- BioHPC – astrocyte (only for biohpc users)
- Open to other ideas!

Independent projects – Present at 4pm!

- Layer on complexity
- Aim for a functional app, even if simple
- Optional to deploy
- Presentations: walk through the app and tell us about next steps in less than 5 min!

